

Joel Harder

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Skills

Languages: JavaScript & TypeScript, Python, PostgreSQL, Dart, Java, C, C++

Technologies: React.js, Vite, Express.js, FastAPI, Next.js, OpenGL, Flutter

Development Tools: Git, Docker, Postman, AWS, Jest, Nginx, Android Studio

Experience

Scotiabank, Data/Software Engineer Intern – IB Treasury Jan 2025 - Aug 2025

- Replaced error-prone manual spreadsheet intake with a full-stack web portal (FastAPI and React), cutting validation time from hours to seconds and standardizing the reporting across 8+ countries.
- Containerized the application with Docker and implemented an Azure DevOps pipeline to publish images to Artifactory, ensuring consistent environments across development and production servers.
- Automated the reporting process from data cleaning to Power BI dashboards, eliminating manual work and improving reliability.

Tulip Retail, Software Developer Co-op Sept 2023 - Dec 2023

- Built custom ticketing tools that automatically generated tickets when errors occur and alert the team via Mattermost, saving 3+ hours of manual reporting each week.
- Improved forms in version 2.0 of the company's web portal (React, TypeScript), simplifying user experience while providing more customizability and reducing friction for non-technical users.

Projects

Study Buddy

🔗 [Read more](#) 🐙 [Github](#)

- Developed an AI-driven web app to instantly turn notes into personalized practice tests, helping users study more efficiently and share tests with others.
- Optimized backend to improve performance and reliability, reducing load times by ~40% and supporting smooth usage during high traffic.
- Tools Used: React.js, Python (FastAPI), PostgreSQL

Chaos Kitchen

🔗 [Read more](#) 🐙 [Github](#)

- Engineered a real-time 2-player mobile game using WebSockets for game state synchronization and Protobufs to cut payload sizes by ~60% vs JSON.
- Designed a 2.5D rendering scheme using layered sprites, achieving 3D visual depth while maintaining 60 FPS on all devices.
- Tools Used: Flutter (Flame Engine), Python (FastAPI), WebSockets, Protobuf, Blender

Cloud-Based VoIP App

🐙 [Github](#)

- Engineered a cross-platform VoIP app and deployed a containerized Node.js backend to AWS using ECS Fargate and an Application Load Balancer, while securely isolating the RDS database to stay under a strict \$50 budget limit.
- Tools Used: React Native, Node.js, Docker, AWS (ECS, ALB, RDS)

Education

BComp. **University of Guelph**, Computer Science with Distinction

Sept 2021 - May 2026

- **Coursework:** Data Structures & Algorithms, Operating Systems, Databases, Networks & Security, Cloud Computing, Mobile Computing, Computer Graphics, Computational Theory